Computer Networks and Internets with Internet Applications, 4e

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Chapter 35

World Wide Web Pages And Browsing

Topics Covered

- 35.1 Introduction
- 35.2 Browser Interface
- 35.3 Hypertext And Hypermedia
- 35.4 Document Representation
- 35.5 HTML Format And Representation
- 35.6 Example HTML Formatting Tags
- 35.7 Headings
- 35.8 Lists
- 35.9 Embedding Graphics Images In A Web Page
- 35.10 Identifying A Page

Topics Covered (cont.)

- 35.11 Hypertext Links From One Document To Another
- 35.12 Client-Server Interaction
- 35.13 Web Document Transfer And HTTP
- 35.14 Browser Architecture
- 35.15 Optional Clients
- 35.16 Caching In Web Browsers
- 35.17 HTTP Support For Caching
- 35.18 Alternative Transfer Protocols
- 35.19 Other Markup Languages

35.1 Introduction

This chapter

- discuss the World Wide Web
- describes the hypertext model and general concepts
- examines the structure of a browser.
- shows how a browser becomes a client, contacts a server, and then displays the information
- explains the structure of a URL
- shows how a brower uses a URL to determine actions

35.2 Browser Interface

- The World Wide Web (WWW) is the large-scale, online repository of information
- A browser is an interactive program used to access the Web
- A browser implements a point and click interface
- A browser can display both text and graphics.
- Some of the displayed information is selectable
  - when the user places the cursor over a selectable item and clicks a mouse button
  - the browser displays new information that corresponds to the selected item

35.3 Hypertext And Hypermedia

- Web is a “distributed” “hypermedia” system that supports interactive access
  - The term “hypermedia” is used to distinguish the Web from a traditional hypertext system which restricts documents to textual information
    - the Web allows documents to contain digitized photographic images or graphics
  - The term “distributed” implies that the Web spans multiple computers, has an important consequence
    - links among Web documents are not guaranteed to remain valid and consistent
    - For example, if a document on one computer contains a link to document D on another computer
    - the link can remain even if document D is removed

35.4 Document Representation

- A hypermedia document available on the Web is called a “page” the main page for an organization or an individual is known as a “homepage”
- Web uses a standard representation
  - known as the HyperText Markup Language (HTML)
- HTML is classified as a “markup language”
  - because it only gives general guidelines for display
  - but does not include detailed formatting instructions
- HTML extensions have been created that do allow the specification of an exact text size
- Markup language is important because it allows a browser to adapt the page to the underlying display: Thus, the same page can be formatted for
  - a large or small computer monitor
  - a high resolution or low resolution display
  - or a handheld device such as a PDA

35.5 HTML Format And Representation

- Each HTML document is divided into
  - a head
    - that contains information about the document
  - a body
    - that contains the information to be displayed
- Syntactically, an HTML document consists of a text file that contains markup tags embedded in the document
- Spaces can be inserted to the source to make it more readable
  - have no effect on the formatted version that a browser displays
  - Figure 35.1 illustrates the general form of an HTML document
  - Document in Figure 35.2 is equivalent to the one in Figure 35.1
- HTML tags provide structure for the document as well as formatting
- Some tags are used to specify a formatting choice that applies to a block of text

```
<HTML>

  <HEAD>
      <TITLE>
            text that forms the document title
      </TITLE>
  </HEAD>

  <BODY>
       body of the document appears here
  </BODY>

</HTML>
```

Figure 35.1 The general form of an HTML document. The head contains information about the document; the body contains the document itself.

```
<HTML><HEAD><TITLE>text that forms
the document
title</TITLE></HEAD><BODY>body of the
document appears here</BODY></HTML>
```

Figure 35.2 The HTML document from Figure 35.1 with some of the unnecessary white space removed. A browser produces the same output for both documents.

35.6- 35.9 Basic HTML Tags

- These sections deal with available tags for
  - Heading
  - Listing items (ordered, unordered)
  - Inserting images
  - Inserting links

35.10 Identifying A Page (1)

- When a user invokes a browser, the user must specify an initial page to view
- Identifying a page is complicated for several reasons:
  - Web includes many computers
    - and a page can reside on any of them
  - A given computer can contain many pages
    - each must be given a unique name
  - Web supports multiple document representations
    - so a browser must know which representation a page uses
  - Because the Web is integrated with other applications
    - a browser must know which application protocol must be used

35.10 Identifying A Page (2)

- The syntactic form encodes the information in a character string known as a
- Uniform Resource Locator (URL)
- The general form of a URL is:

  protocol://computer_name:port/document_name
- where
  - protocol → the name of the protocol used to access the document
  - computer_name → the domain name of the computer
  - port → an optional protocol port number
  - document_name → the name of the document
- Because the optional protocol port number is seldom used
  - we will omit it from the discussion
- For example, the URL

  http://www.netbook.cs.purdue.edu/toc/toc01.htm

35.11 Hypertext Links From One Document To Another

- Each hypertext reference is a passive pointer to another document
  - a hypertext reference does not cause an immediate action
  - Instead, a browser turns a hypertext reference into a selectable item when displaying the document
  - If the user selects the item
    - the browser follows the reference, retrieves the document to which it refers, and replaces the current display with the new document
- HTML allows any item to be designated as a hypertext reference
  - Thus, a single word, a phrase, an entire paragraph, or an image can refer to another document

35.12 Client-Server Interaction (1)

- When given the URL of a document
  - a browser becomes a client that contacts a server on the computer specified in the URL to request the document
- The browser then displays the document for the user
- Unlike network applications that arrange for a client and server to maintain an established connection
  - the TCP connection between a Web browser and a server has a short duration
    - The browser establishes a connection
    - sends a request
    - and receives the requested item or a message that no such item exists
    - as soon as the document or image has been transferred, the connection is closed
    - the client does not remain connected to a server.

35.12 Client-Server Interaction (2)

- Terminating connections quickly works well in most instances
  - Because browsing does not exhibit high “locality”
  - A user might access a Web page on one computer, and then immediately follow a link to a Web page on another computer
- But, terminating connections too quickly can introduce overhead
  - in cases where a browser must return to the same server for many documents
  - For example, consider a page that contains references to several images, all of which reside on the same computer as the page
- When a user selects the page, the user's browser opens a connection, obtains the page, and closes the connection
  - When it needs to display an image the browser must open a new connection to the same server to obtain a copy of the image
- Thus, new version of the protocol allow a browser to retain a single TCP connection across multiple requests (persistant)

35.13 Web Document Transfer And HTTP (1)

- When a browser interacts with a Web server, the two programs follow the
  - HyperText Transfer Protocol (HTTP)
- In principle, HTTP is straightforward
  - it allows a browser to request a specific item
  - which the server then returns
- In practice, HTTP is complex because each side can send additional information
  - that specifies the status and contents of the message

35.13 Web Document Transfer And HTTP (2)

HTTP supports four basic operations that a browser can specify when making a request:

- GET → requests a specified item from the server
  - The server returns a heading that contains status information followed by a blank line followed by the item
- HEAD → requests status information about an item
  - The server returns the status without returning a copy of the item itself
- POST → sends data to the server
  - The server appends the data to a specified item
- PUT → sends data to the server
  - The server uses the data to replace a specified item .
  - It is seldom used; most Web pages that send data use POST instead

```
HTTP/1.0 200 OK
Date: Mon,  1 Mar 2004 01:22:22 GMT
Server: Apache/1.2.5
Last-Modified: Thu, 30 Oct 2003 10:03:37 GMT
ETag: "130fe-81-3883bbe9"
Content-Length: 129
Accept-Ranges: bytes
Connection: close
Content-Type: text/plain
```

Figure 35.4 Example HTTP header returned by a server. The status code *200* on the first line indicates that the server honored the request; additional lines give further information about the requested item.

35.14 Browser Architecture

- Web browsers have a more complex structure than Web servers:
  - A server performs a straightforward task repeatedly
  - the server waits for a browser to open a connection and request a specific page
    - The server then sends a copy of the requested item, closes the connection, and waits for the next connection
- A browser handles most of the details of document access/display
  - Consequently, a browser contains several large software components that work together to provide the illusion of a seamless service.
- A browser consists of
  - a set of clients
  - a set of interpreters
  - and a controller that manages them
- The controller forms the central piece of the browser.
  - It interprets both mouse clicks and keyboard input
    - and calls other components to perform operations specified by the user
- Figure 35.5 illustrates the conceptual organization of a Web browser

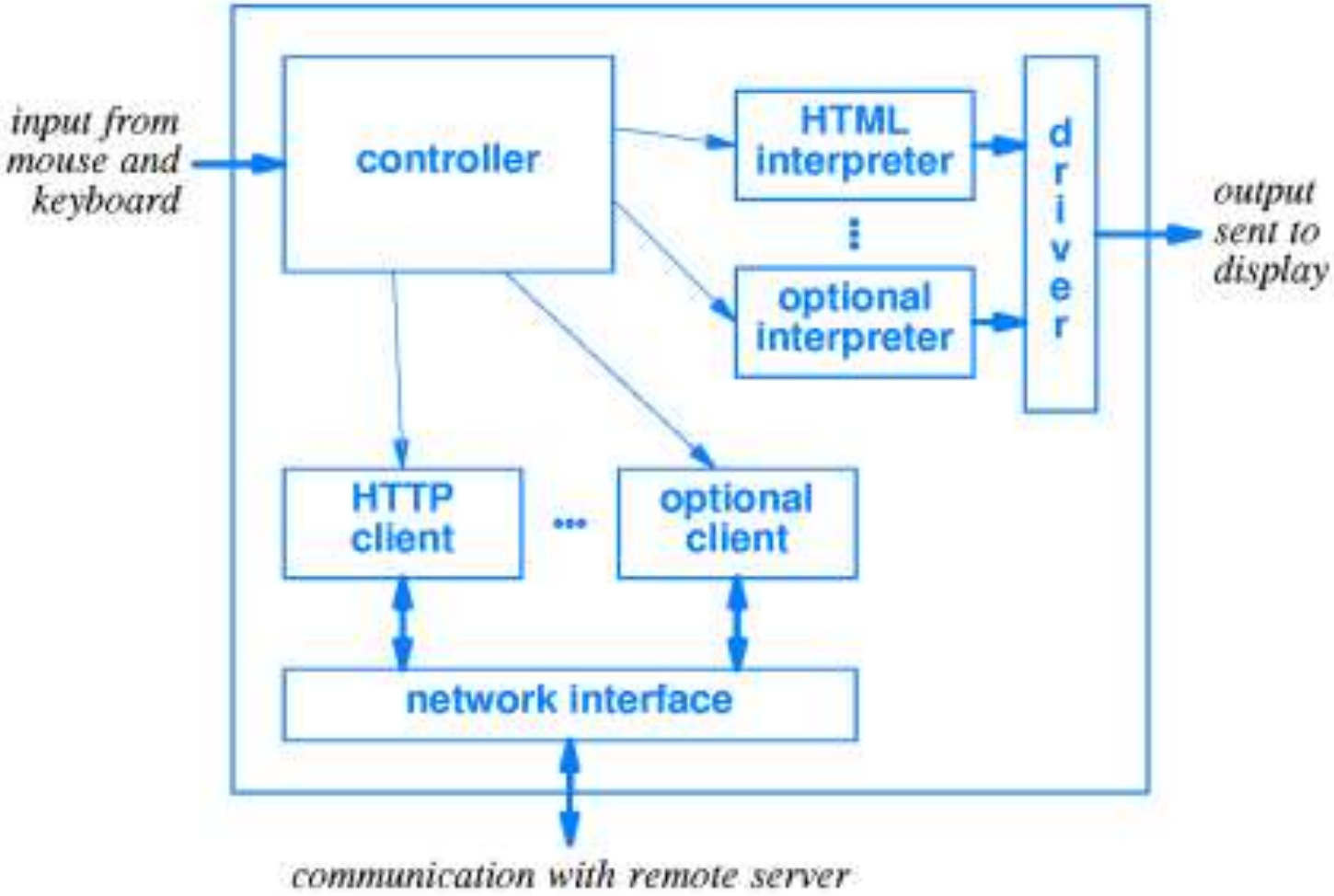

Figure 35.5 Major components of a Web browser. Dark arrows show the flow of data; other arrows show control paths. The data paths from clients to interpreters are not shown.

35.15 Optional Clients

- Abrowser can contain components that enable the browser to perform additional tasks
  - Exp: many browsers include an FTP client
  - Some may contain e-mail client software
- How can a user access a service such as FTP from within a browser?
  - The answer is that a user does not invoke such services explicitly nor does the user interact with conventional client software
    - Instead, the browser invokes the service automatically, and uses it to perform a task as needed
- If the browser is well-designed
  - it hides details from a user, who may be unaware that an optional service has been invoked
  - For example, file transfer can be associated with a selectable item on the screen
    - When the user selects the item, the browser uses its FTP client to obtain a copy of the file
- How can file transfer be specified in a Web page?

  ftp://ftp.cs.purdue.edu/pub/comer/netbook/client.c

35.16 Caching In Web Browsers (1)

The pattern of client-server contact in Web browsing is unlike the pattern in many of the applications described earlier, because

- users tend to view Web pages outside of their organization
  - Browsers tend to reference remote pages more frequently than local pages
- users do not search for the same information repeatedly
  - they do not tend to repeat accesses day after day

- Because browsing produces a different “locality of reference” than other applications
  - browsers use to optimize performance differ from other applications
  - neither browsers nor Web servers are optimized for physical locality
  - furthermore, browsers employ techniques to handle “temporal locality of reference”

35.16 Caching In Web Browsers (2)

- The browser places a copy of each item it retrieves in a cache on the local disk
  - When a user selects an item, the browser checks the disk cache before retrieving a copy
  - If the cache contains the item
    - the browser obtains the copy from the cache
  - if the page cannot be found in the cache
    - it contacts the origin server
- Keeping items in a cache can improve performance dramatically
  - a browser can read the item from disk without waiting for network delays
  - it is especially important for large pages or for users who have slow network connections
- Despite the large improvement in speed, retaining items in a cache for extended periods is not always desirable
  - a cache can take vast amounts of disk space.
  - improvements are only helpful if a user decides to view an item again

35.17 HTTP Support For Caching

- HTTP has several features that make caching possible
  - When the server generates an HTTP header
    - the server can specify a cache timeout for the page.
  - Alternatively, if the page contains information that changes constantly
    - the server can specify that the page may not be cached.
  - Similarly, a browser can send an HTTP header that specifies the request must not be answered with data from a cache
- The HTTP HEAD operation also helps with cache management
  - because cache timeout values are usually conservative
  - Instead of downloading a fresh copy
    - the browser can attempt to revalidate the cached version

35.18 Alternative Transfer Protocols

- Several extensions and alternatives have been proposed to enhance or replace HTTP
- HTTPS
  - protocol provides for secure HTTP transfers
    - (i.e., it uses encryption to guarantee confidentiality of the data being transferred in an HTTP session)
- HTTP version 1.1 introduces the concept of persistent connections
  - in which a single TCP connection is used for multiple transfers
    - (i.e., multiple GET requests from the browser).
- HTTP-NG (Next-Generation of HTTP)
  - that will use binary encoding for headers instead of the current ASCII representation

35.19 Other Markup Languages

Other markup languages have been created for special purposes.

- Voice Extensible Markup Language (VoiceXML) or VXML
  - specifies communication in human speech
- Extensible Markup Language (XML)
  - Unlike HTML, XML does not specify layout
  - Instead, XML gives an internal representation that provides names for each field in a data item
  - Figure 35.6 contains an example of XML

```
<ADDRESS>
    <NAME>
        <FIRST>  John    </FIRST>
        <LAST>   Public  </LAST>
    </NAME>
    <OFFICE> Room 320      </OFFICE>
    <PHONE>  765-555-1234 </PHONE>
</ADDRESS>
```

Figure 35.6 An example of XML for a corporate phone book. Each data item is given a name.